

BUSINESS PROCESS MANAGEMENT

1 – Introduction and Foundations

T. Beurskens & C. Salz
Feb 2025

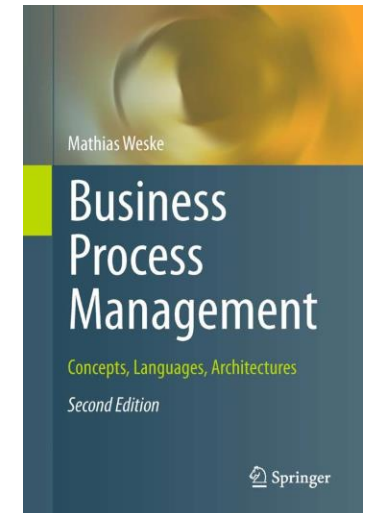
Starter



- Question: A man walks into a restaurant and orders food. When the food arrives, he starts eating. How can this be without paying?
- Answer: He is part of the restaurant's staff and gets a free meal as a work benefit.

Definition – Business Process

- This PowerPoint file contains figures used in
 - Mathias Weske, Business Process Management: Concepts, Languages, Architectures 2012, XIV, 403 p. 300 illus., Hardcover ISBN: 978-3-642-28615-5, eBook ISBN: 978-3-642-28616-2 © Springer-Verlag Berlin Heidelberg 2012, 2007
- This module is partly based on that book



Business Process

- Business Process

consists of a set of activities that are performed in **coordination** in an organizational and technical environment. These activities jointly realize a **business goal**. Each business process is enacted by a single organization, but it may interact with business processes performed by other organisations.

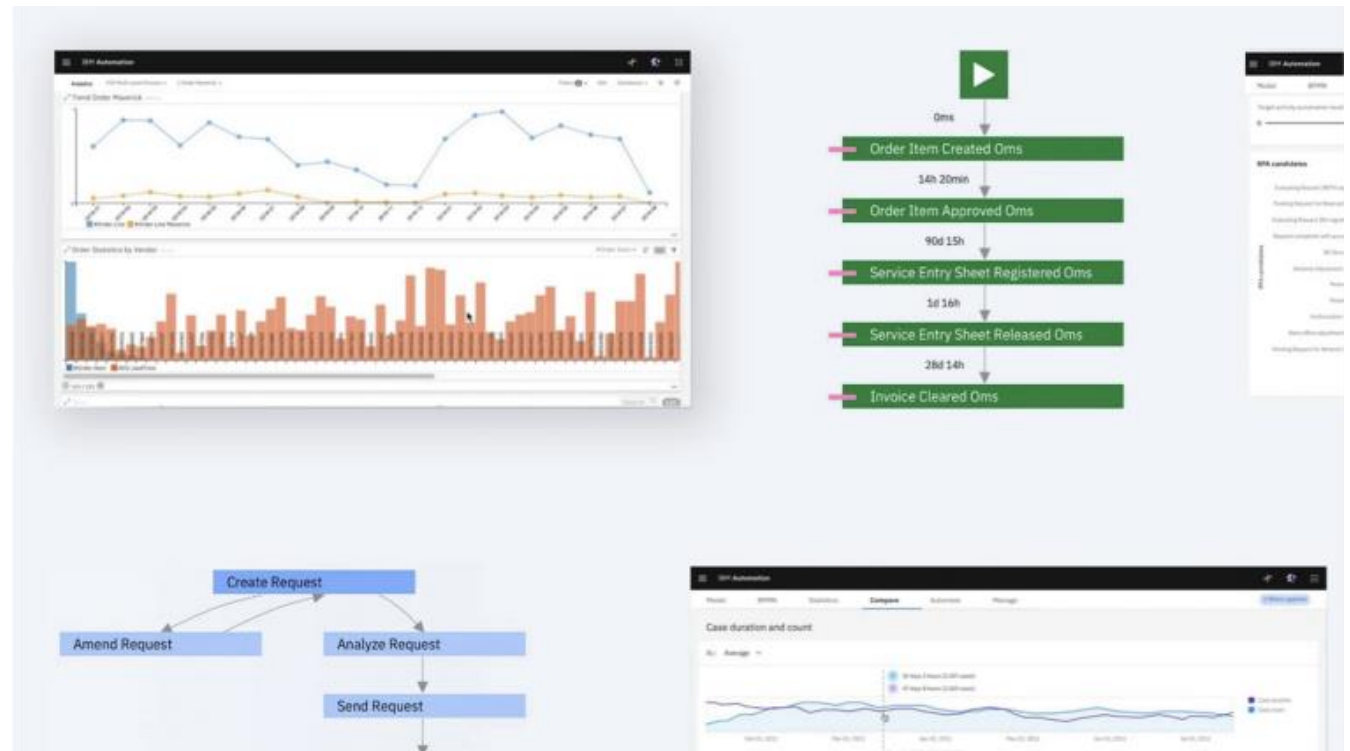
Business Process Management

- Business Process Management

includes concepts, methods and techniques **to support the design, administration, configuration, enactment, and analysis** of business processes.

Business Process Management System

- Business Process Management System is a generic software system that is driven by explicit process representations to coordinate the enactment of business processes.



Simple business process model

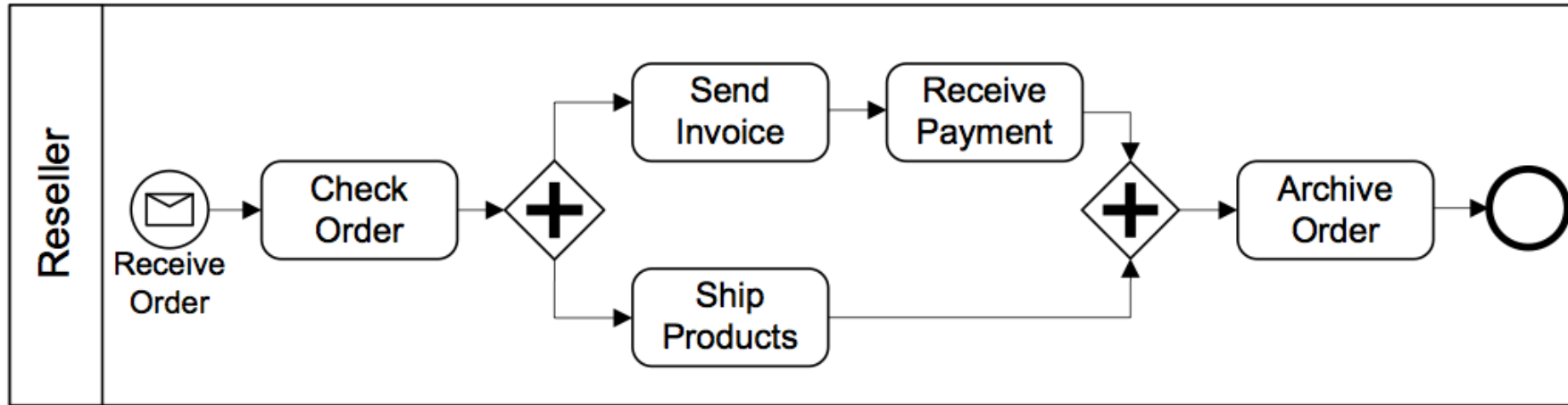


Fig. 1.1. Simple ordering process of reseller

M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

- Activities are represented by rounded rectangles, marked with the name of the activity
- Events can be used to mark the start and end of the process, events are represented by circles.
- Branching and joining of nodes is represented by diamonds that can be marked with different symbols. (parallel split & join)

Process Orchestrations

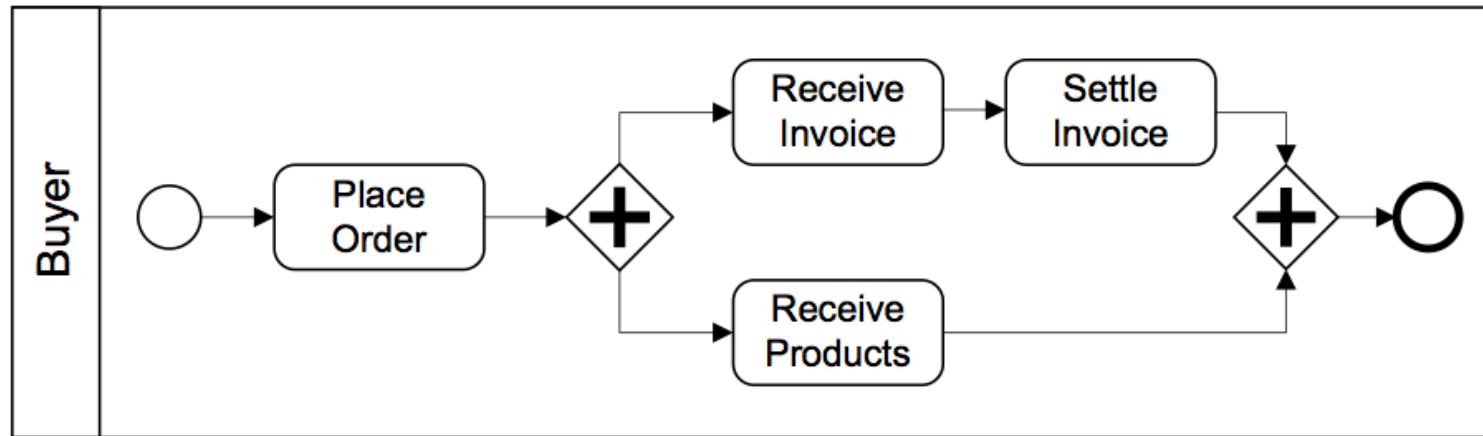
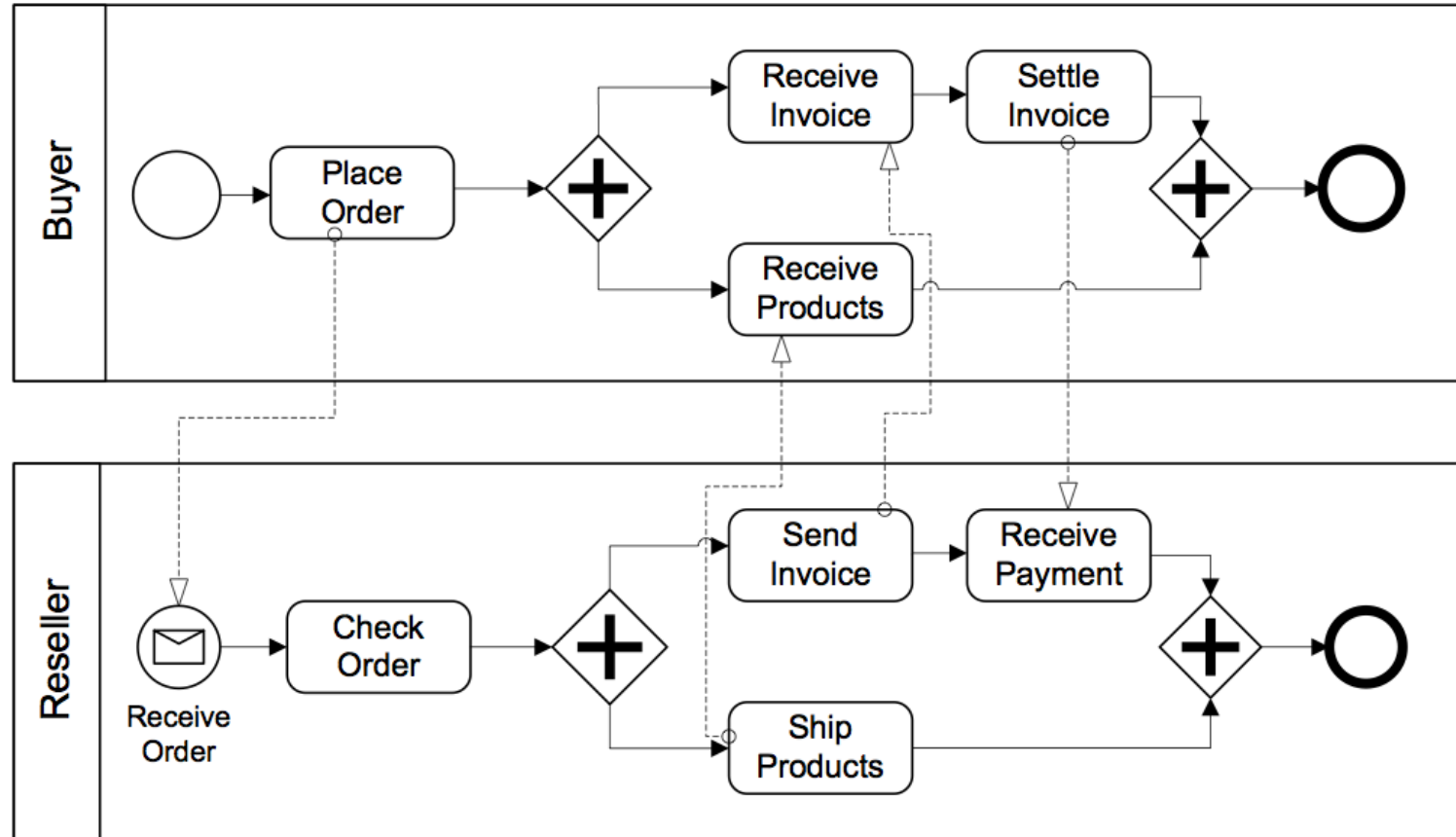


Fig. 1.2. Ordering process of a buyer

M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

- Business processes are performed in a single organisation by definition

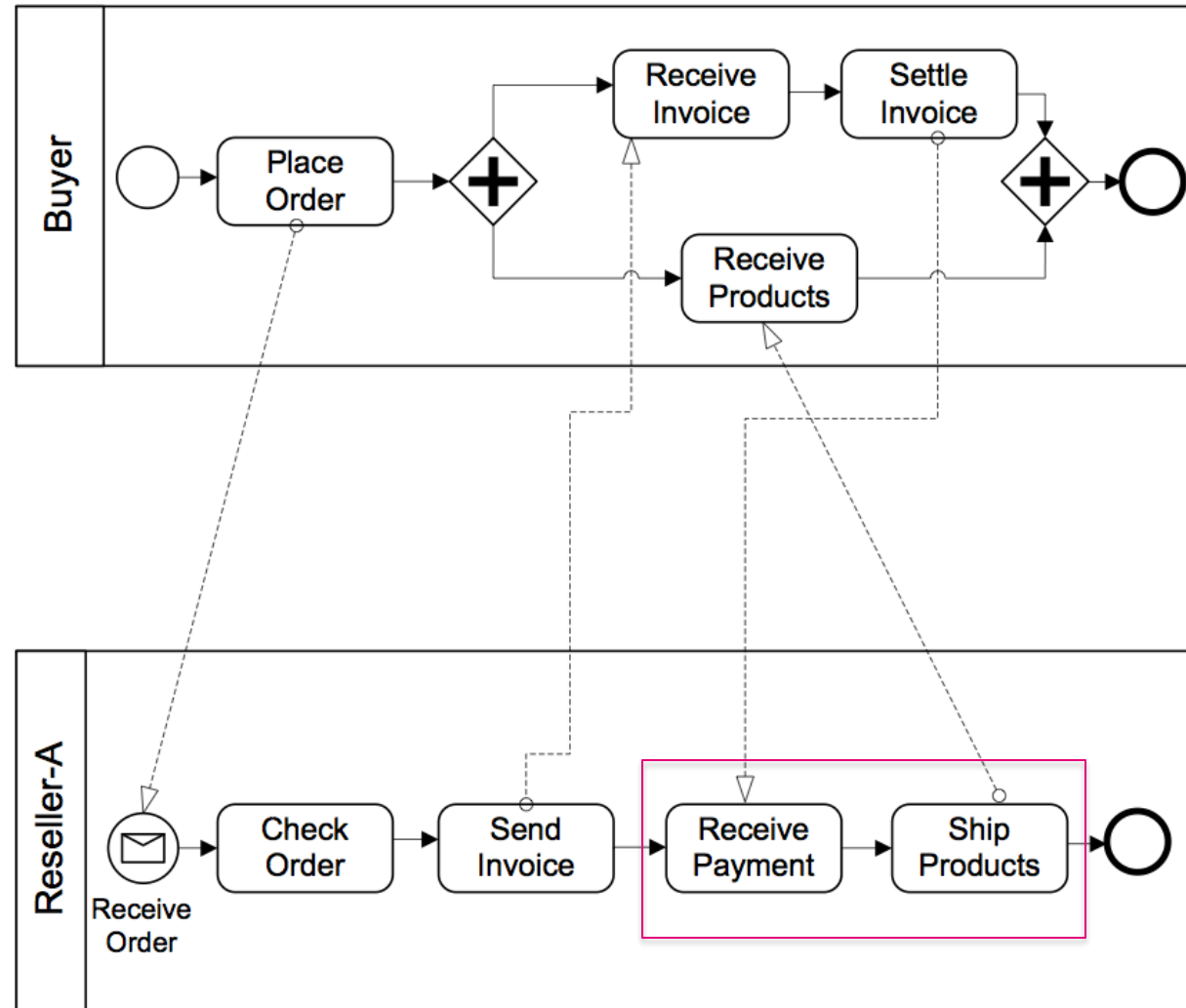
Process Choreographies



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 1.3. Interacting business processes form process choreography

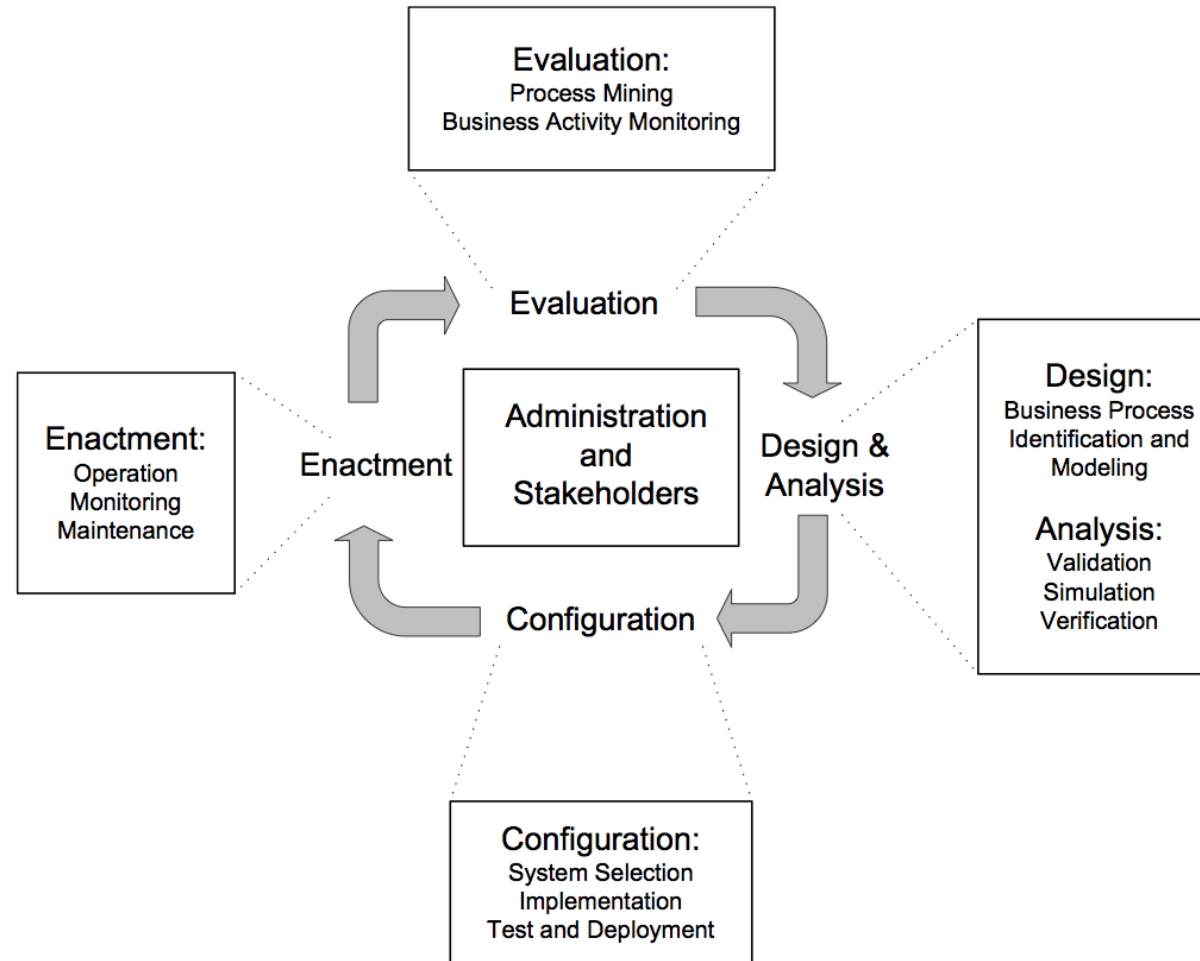
There is no focus on technical aspects of realization



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 1.4. Variant of reseller process with interacting business process

The Business Process Lifecycle



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 1.5. Business process lifecycle

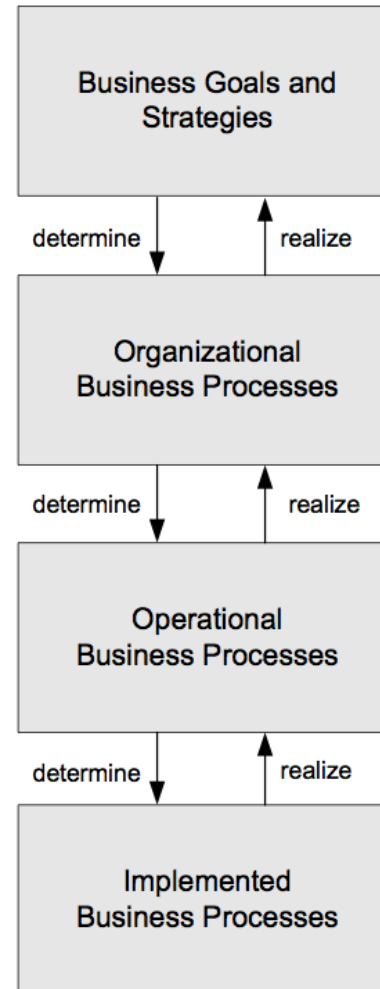
Organisational vs. Operational

- Business goal
 - Long term business objective
- Business strategy
 - Plan for achieving goals
- Organizational business processes
 - Typically in textual form
 - Inputs, outputs, expected results, dependencies
 - Informal and semi-formal techniques are used

Organisational vs. Operational

- Operational business processes
 - Details of organizational process
 - Activities and their relationships are defined
 - Implementation aspects are disregarded
 - Focuses on **what** needs to be done rather than **how** it is executed
 - Defines **activities and their relationships** (e.g., "Customer places an order → Order is processed → Order is shipped").
- Implemented business processes
 - Execution of the process activities
 - Technical and organizational environment
 - Allows enactment on a given platform
 - Specifies **how** a process is executed in a **technical and organizational environment**.
 - **Includes technical details** (e.g., which software system automates each step).

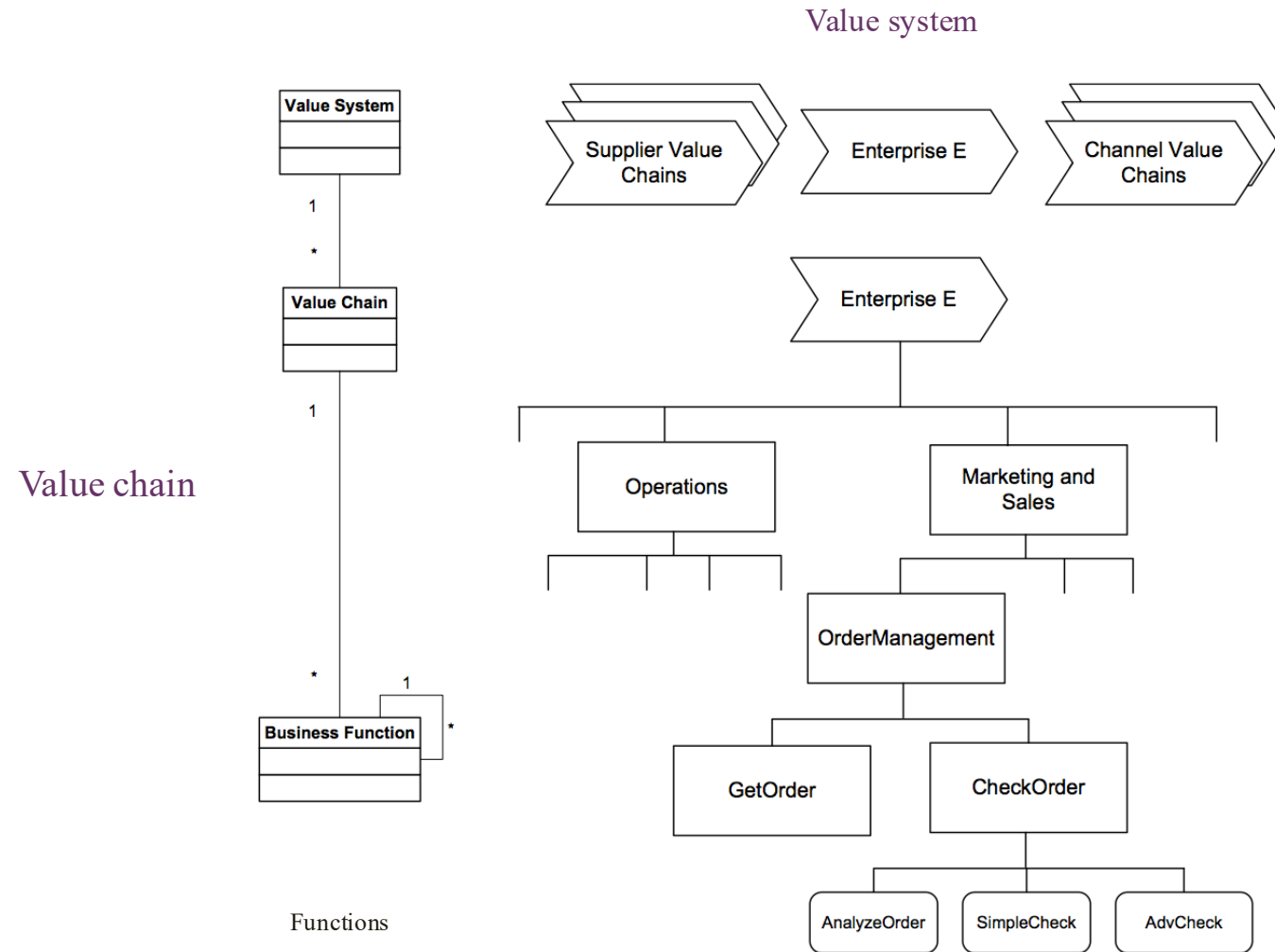
Classification of Business Processes



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 1.6. Levels of business processes: from business goals and strategies to implemented business processes

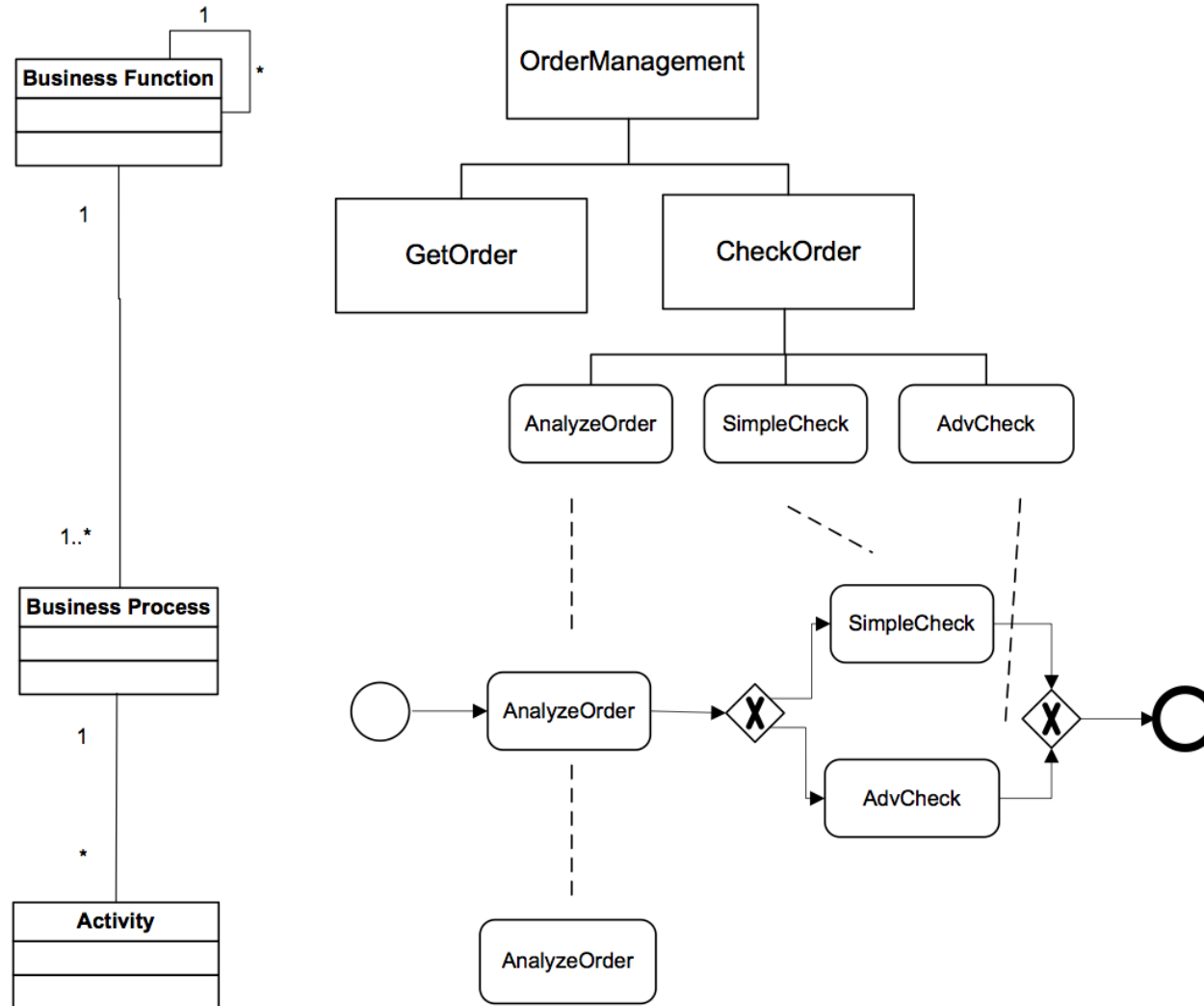
Value system (1/3)



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 3.4. Functional decomposition from value chain to business functions

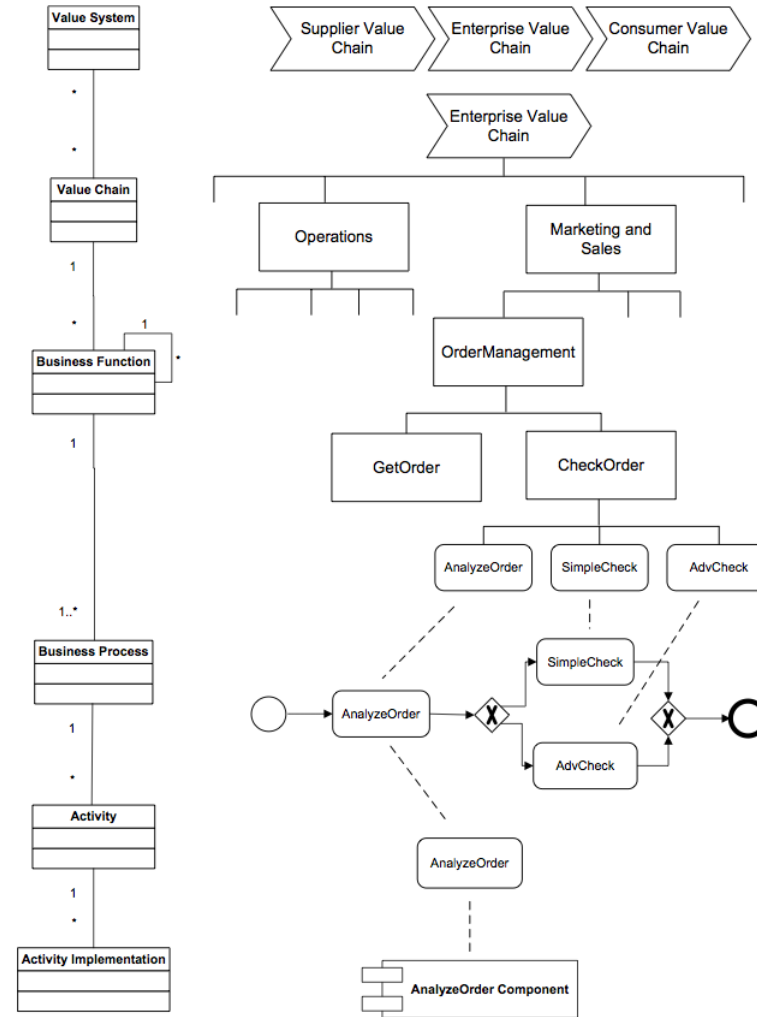
Value system (2/3)



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 3.5. Business functions of small granularity are organized as a business process

Value system (3/3)



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 3.7. Levels of business process management. From value systems to activity implementations

State transition diagram

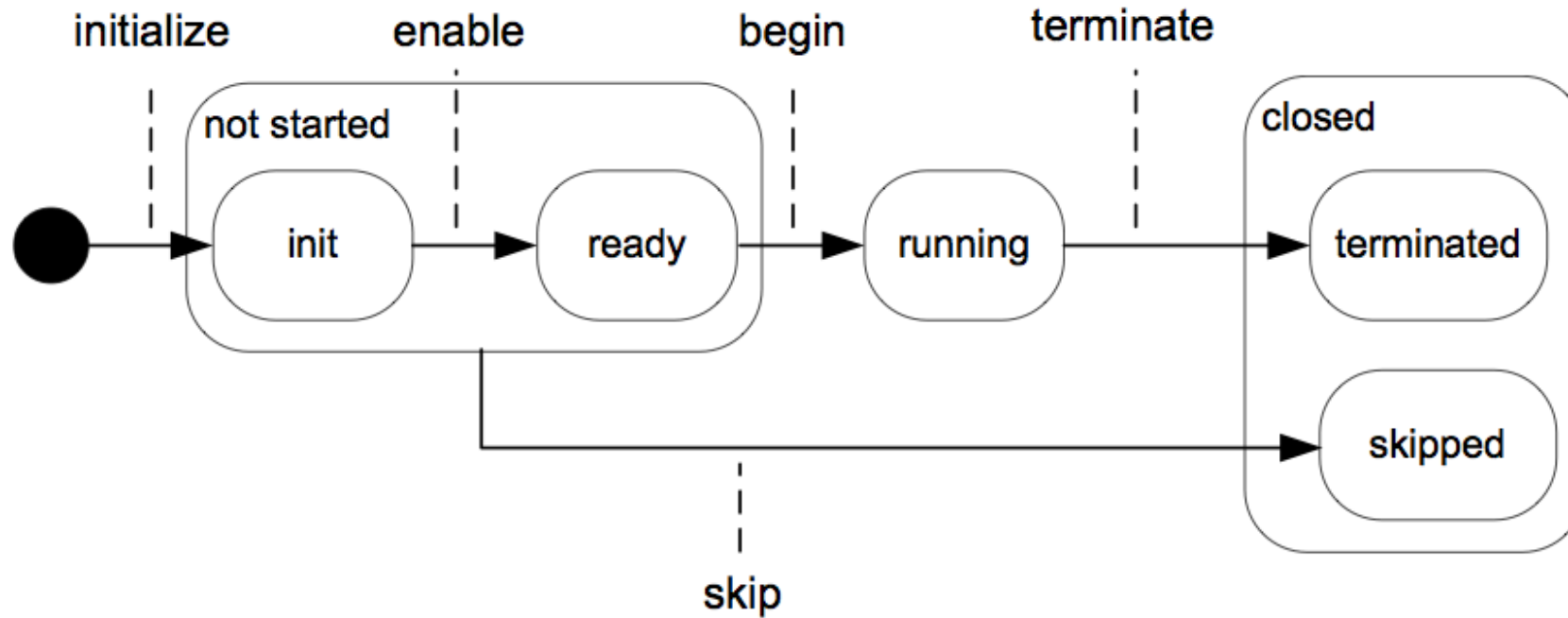
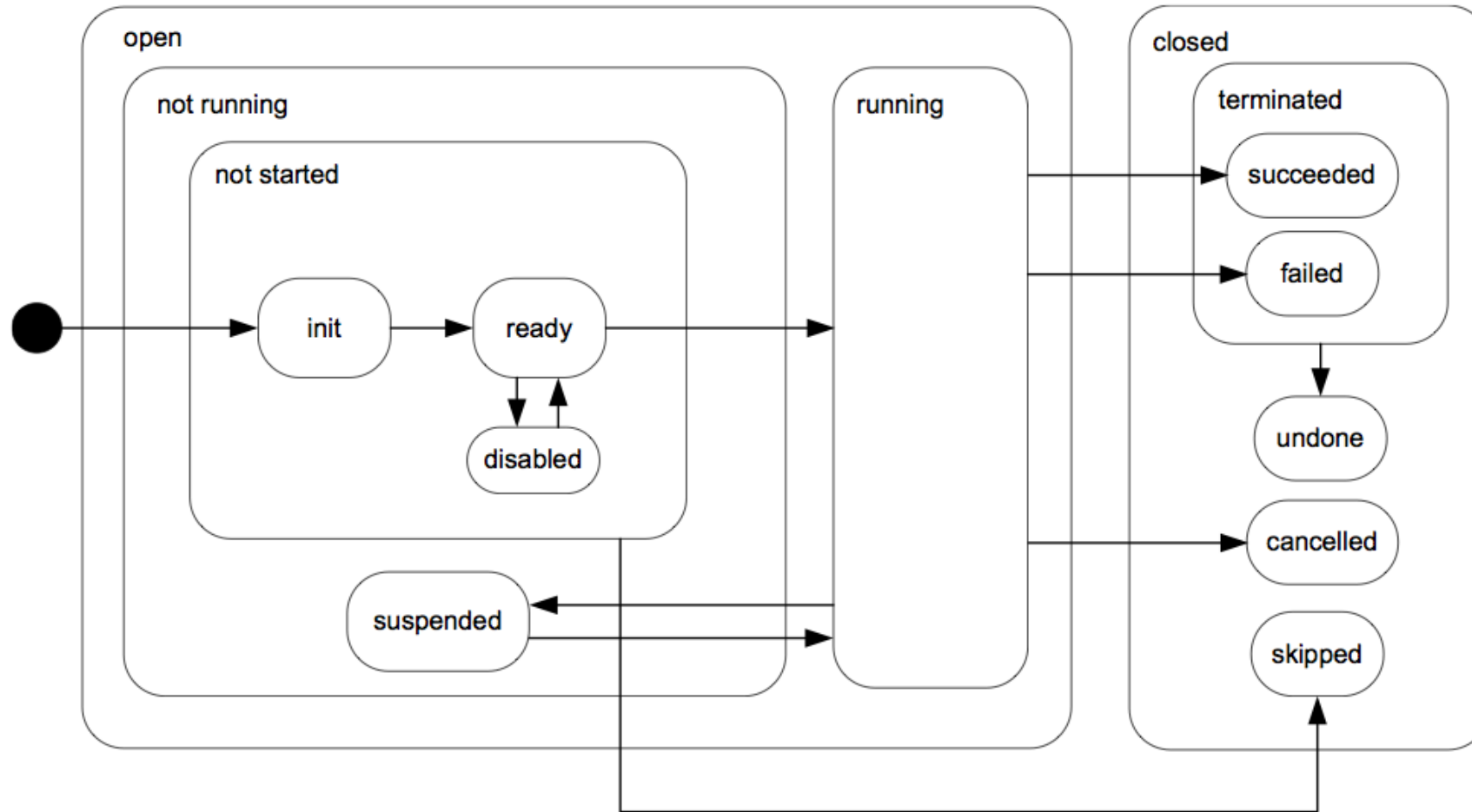


Fig. 3.9. State transition diagram for activity instances

M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

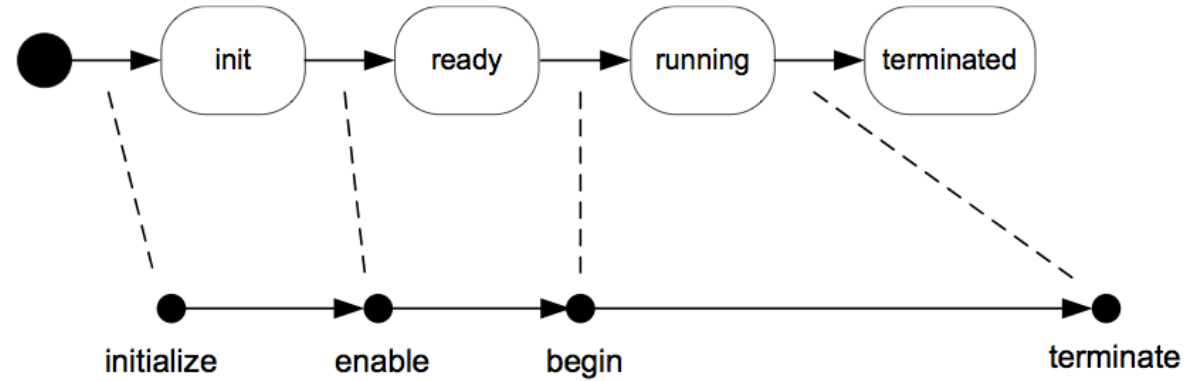
State transition diagram, detailed



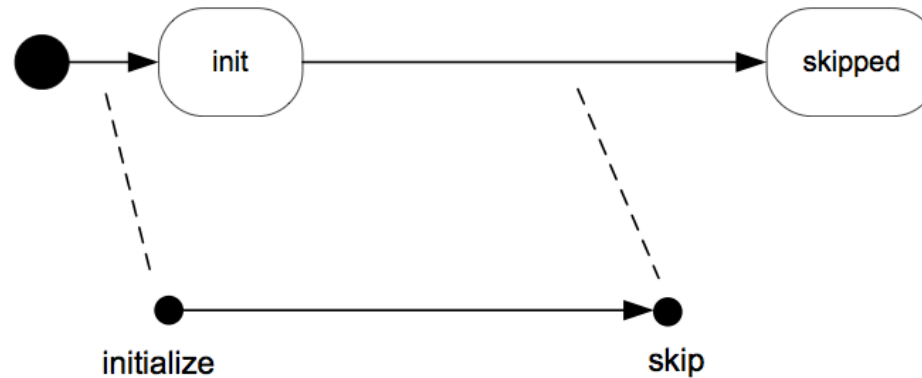
M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 3.10. State transition diagram for activity instances, detailed version

State transition diagram



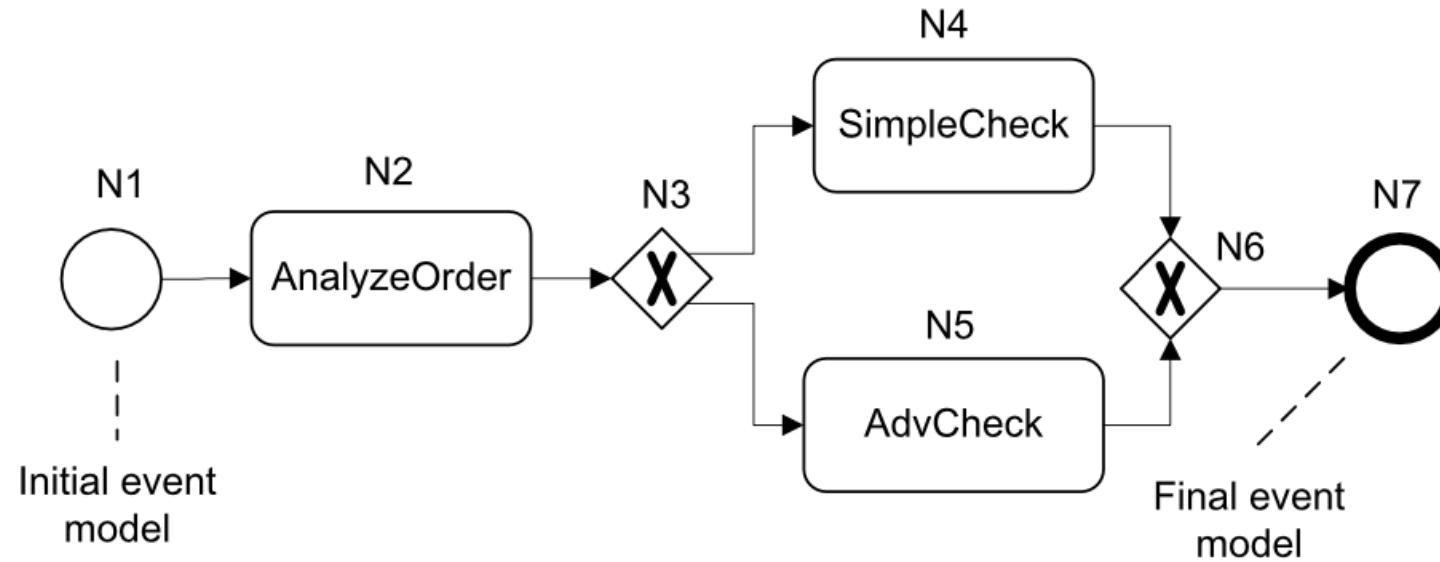
(a)



(b)

Fig. 3.11. Event diagram for (a) executed activity instance and (b) skipped activity instance

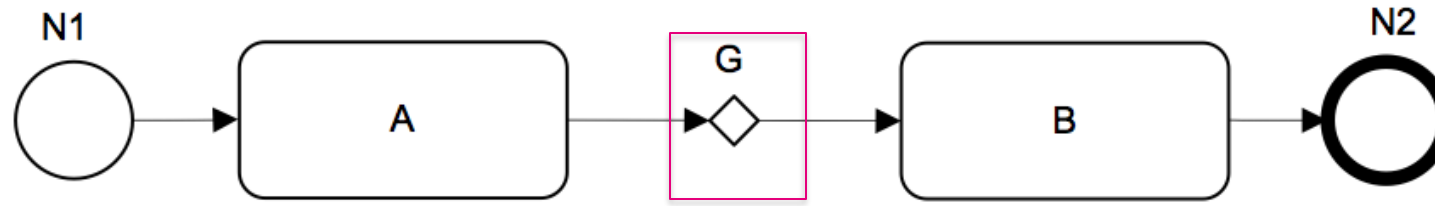
Nodes



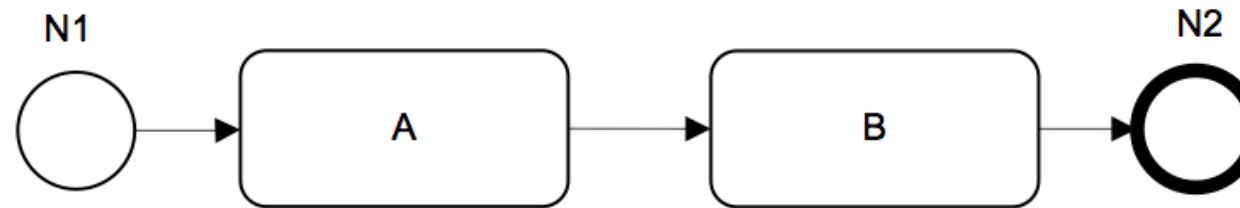
M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 3.14. Process notation used to express concepts from process meta model

Sequence gateway



(a) Sequence Flow with Sequence Gateway



(b) Sequence Flow, Simplified Notation

Fig. 3.15. Alternative representations of sequence gateway

Event diagram (1/2)

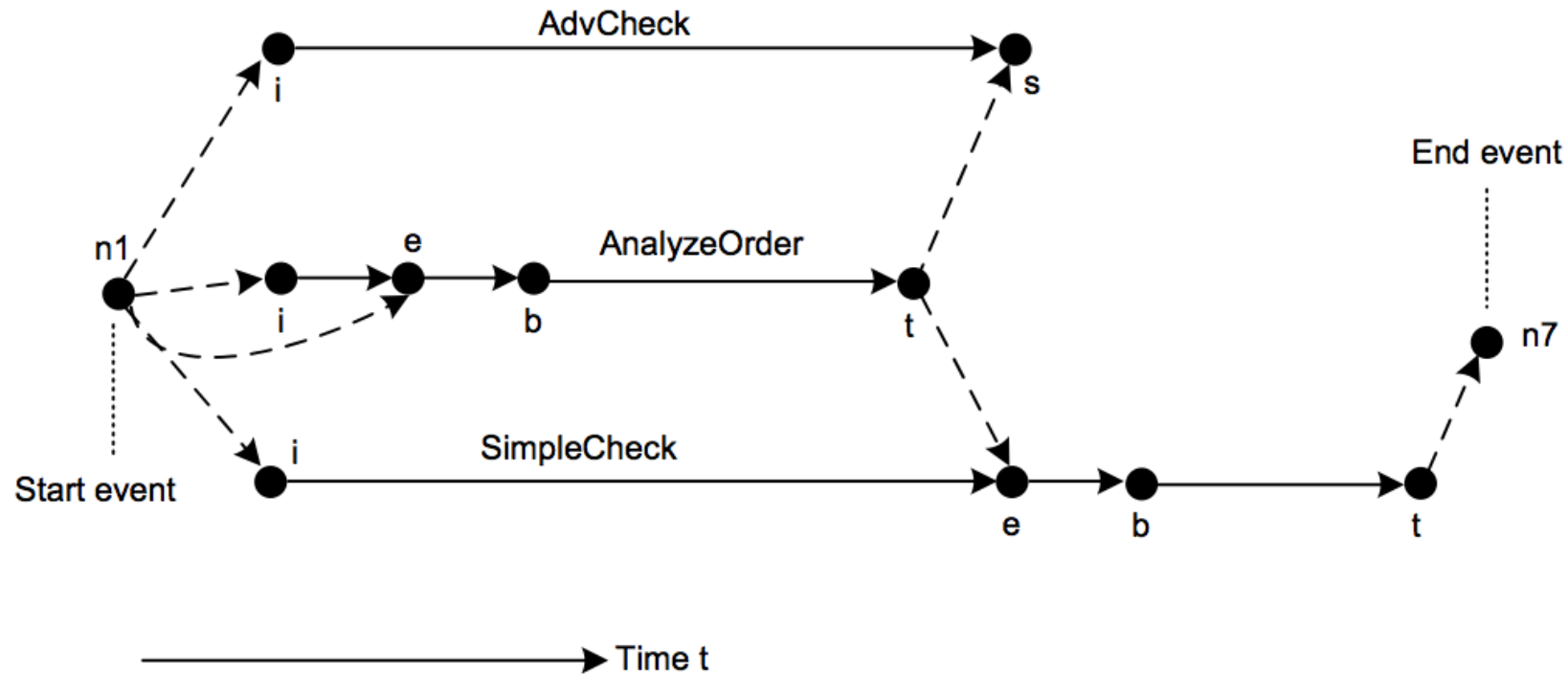


Fig. 3.17. Event diagram of sample process instance (subscripts of *init*, *enable*, *terminate* and *skip* events omitted)

Event diagram (2/2)

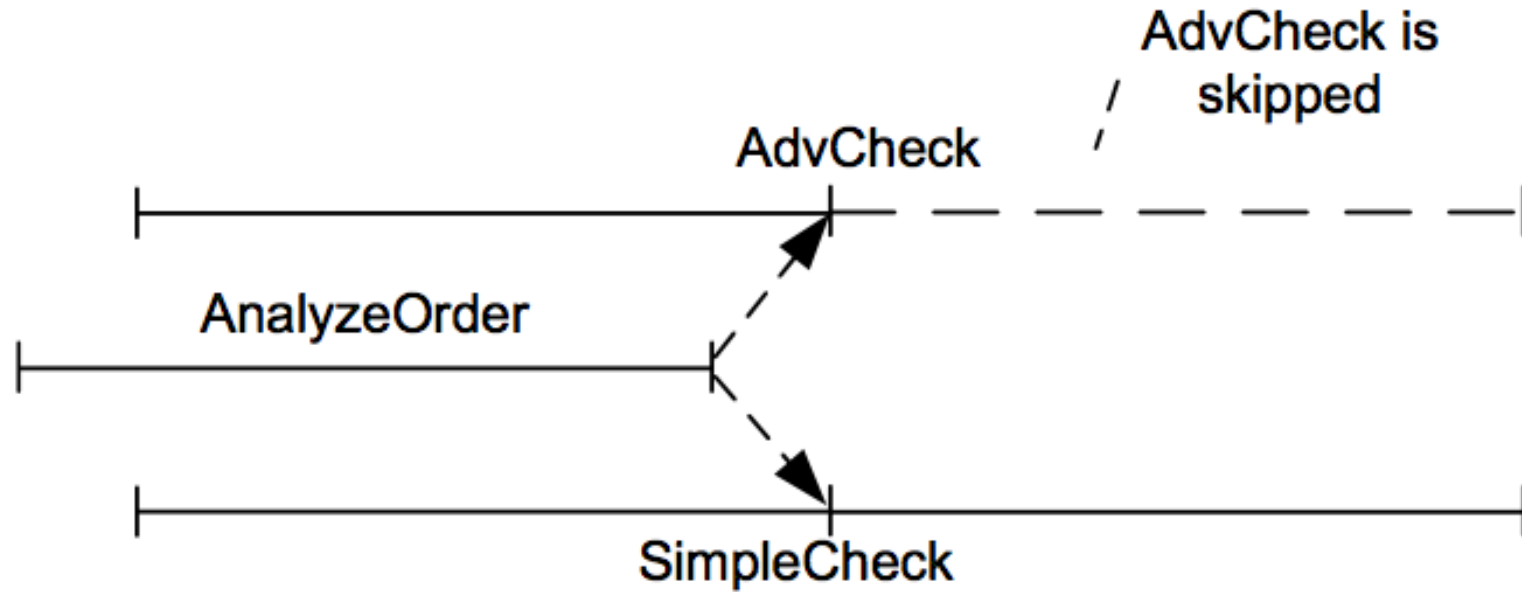
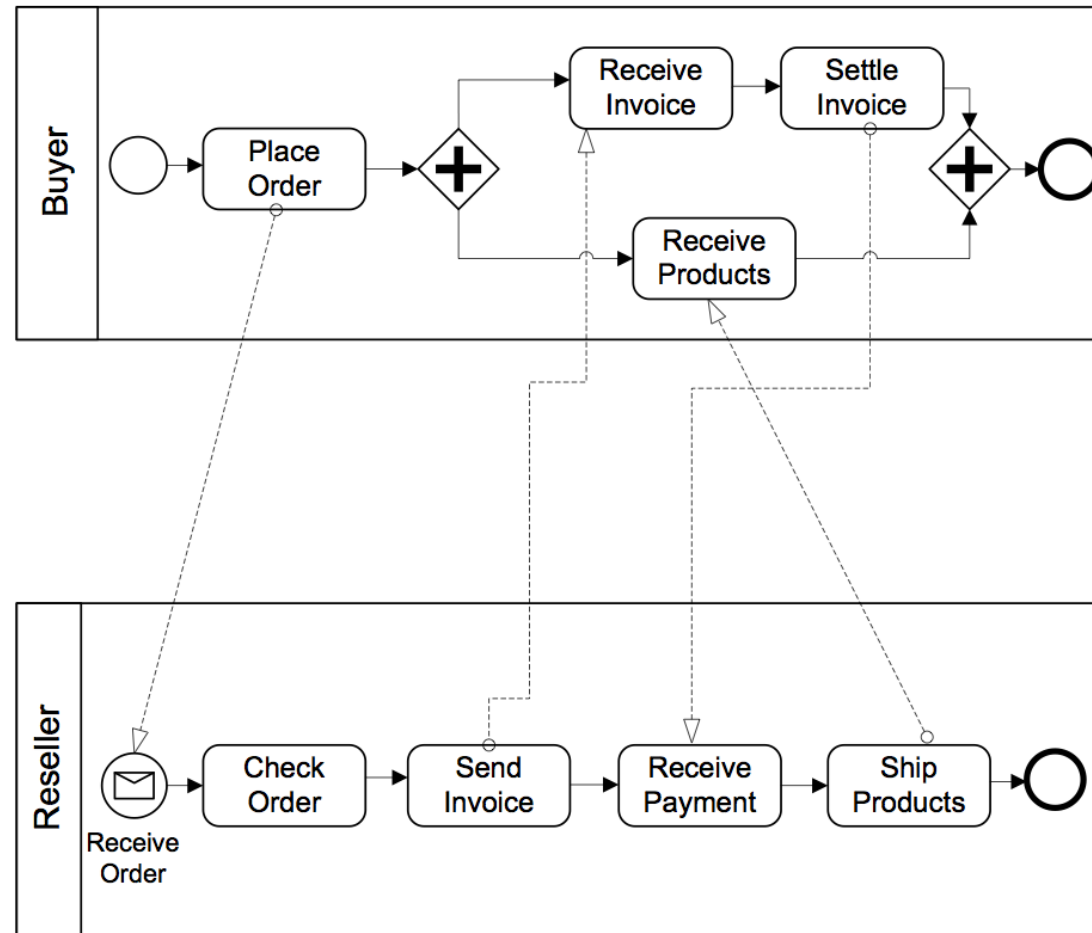


Fig. 3.19. Compact representation of event diagrams

Process Interactions (1/2)



M. Weske: Business Process Management,
 © Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 3.20. Interacting business processes involving buyer and reseller

Process Interactions (2/2)

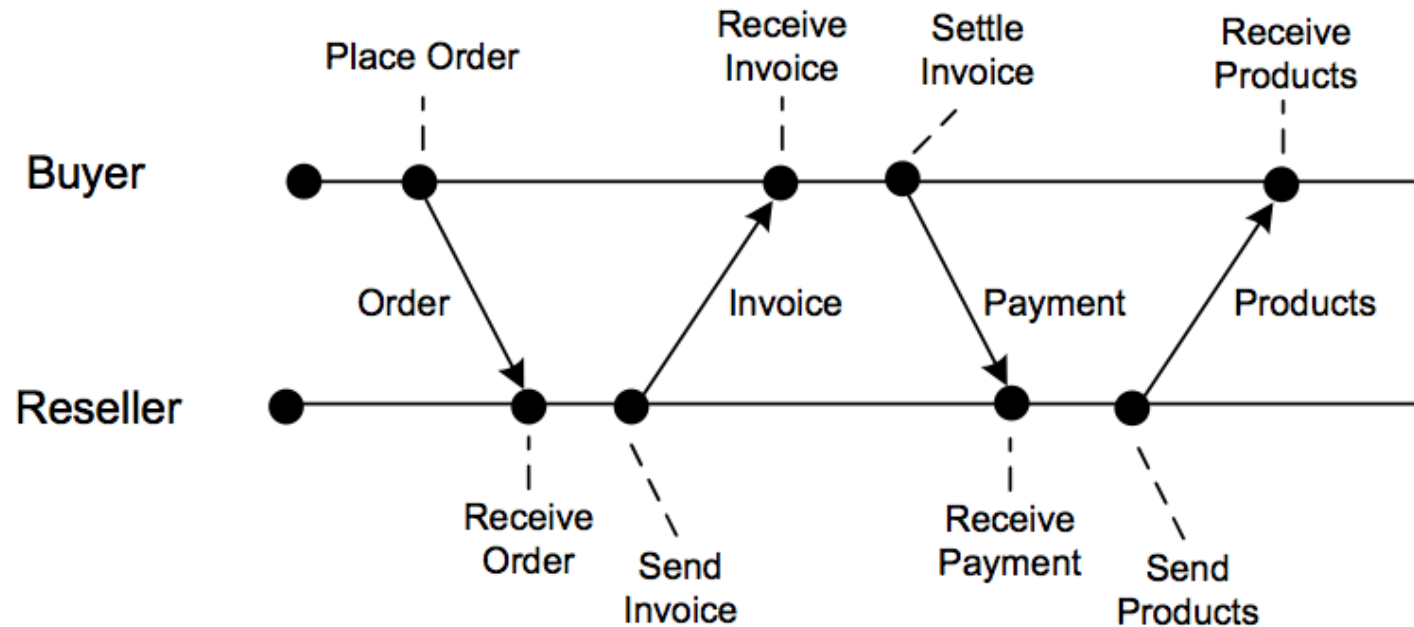
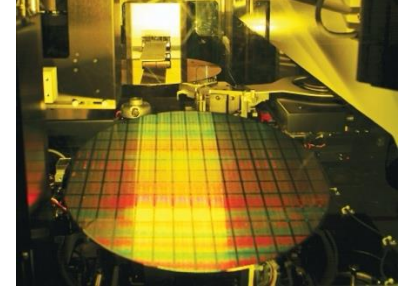


Fig. 3.21. Event diagram of interacting processes shown in Figure 3.20

M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Workflow Data Patterns



- Dimensions: data visibility, data interaction, data transfer, data-based routing
- Task data
 - Data object is local to a particular activity – it is not visible to other activities (of same or of other process) -> Example: CPU thermal paste temperature
- Block data
 - Data object is visible to all activities of a given subprocess -> Example: GPU assembly subprocess data
- Workflow data
 - Data object is visible to all activities of a given business process – access is restricted by business process mgmt. system as defined in business process model -> Example: Production batch ID, defect rates, and production time per unit
- Environment data
 - Data object is part of business process execution environment – it can be accessed by process activities during process enactment -> Example: Factory-wide resource availability, energy consumption

Patterns: Assign participants to activities

- Direct allocation – individual person allocated to all activity instances
- Role-based allocation – role is assigned to each activity model
- Deferred allocation – allocation decision is made at run time of business process
- Authorization – allocation based on a person's position
- Separation of Duties – e.g. document needs to be signed and countersigned
- Case handling – all activity instances handled by one knowledge worker
- History-based allocation – e.g. one-face to the customer; do what you already did previously
- Organizational allocation – similar to authorization; e.g. head of department (position) to work on orders > 50K€

Why BPMN?

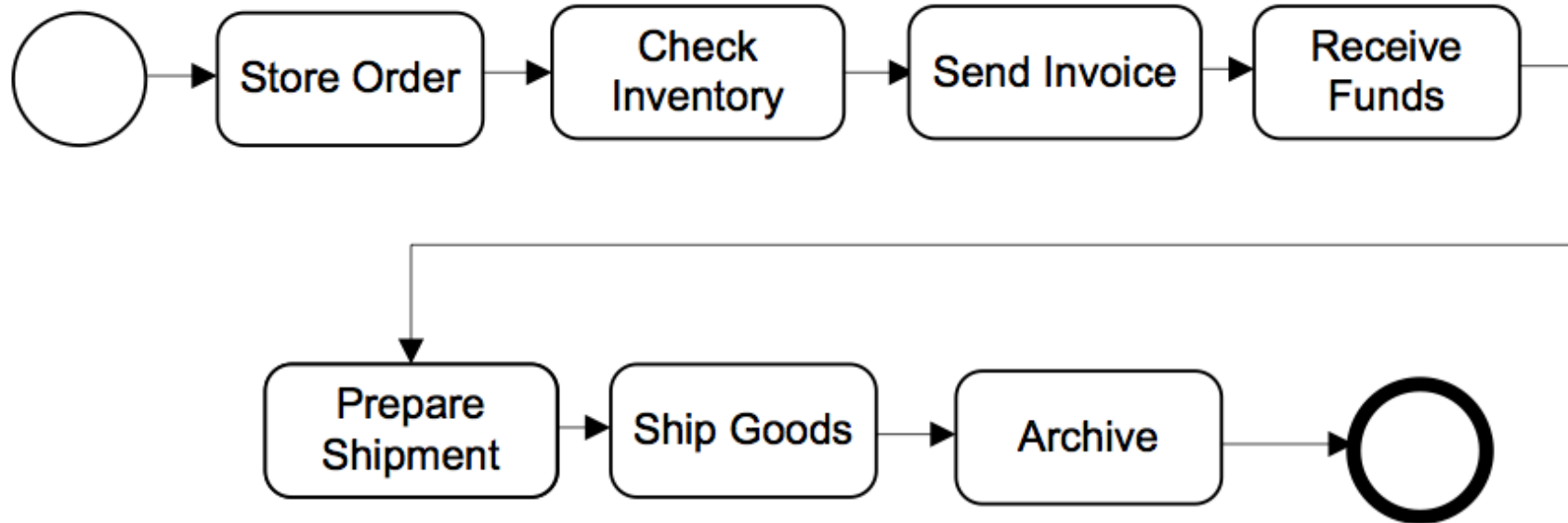


Fig. 3.31. Sample business process model

- This process model might suffer from long processing times, resulting in insufficient customer satisfaction.

Improved business process

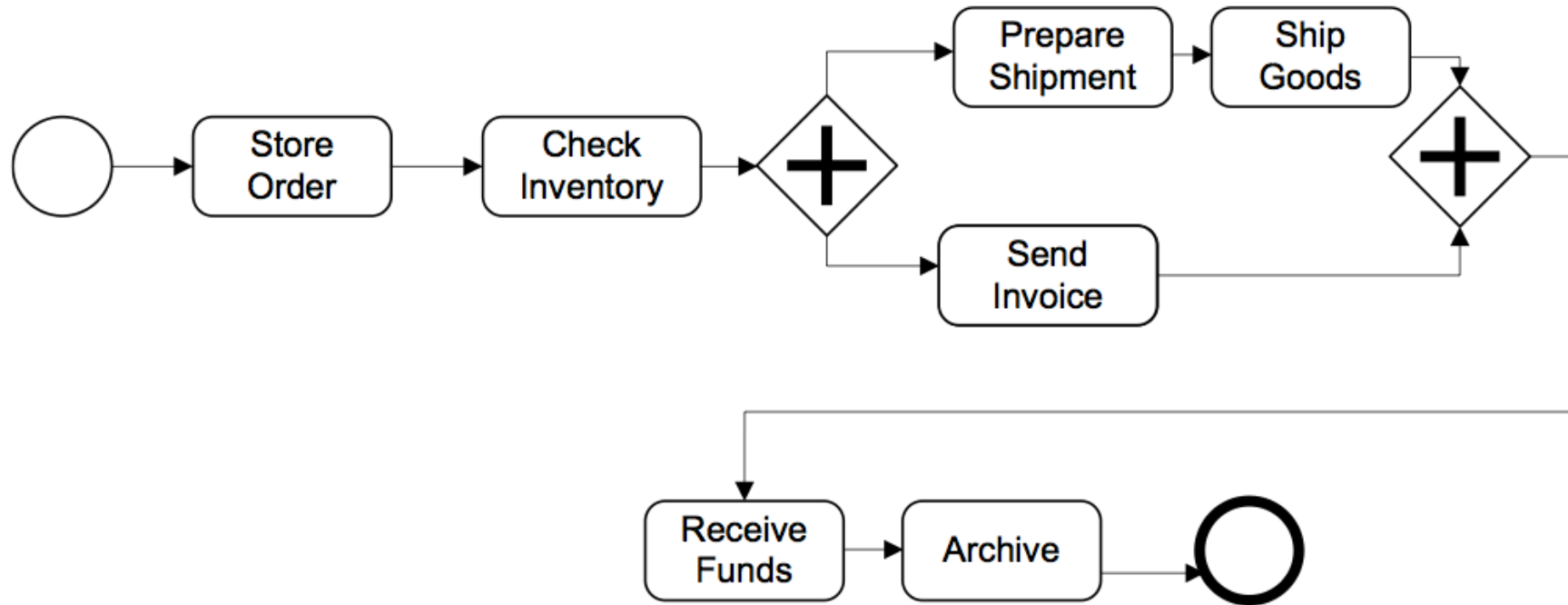


Fig. 3.32. Sample business process model, improved version with concurrency

M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Organizational Modelling (1/3)

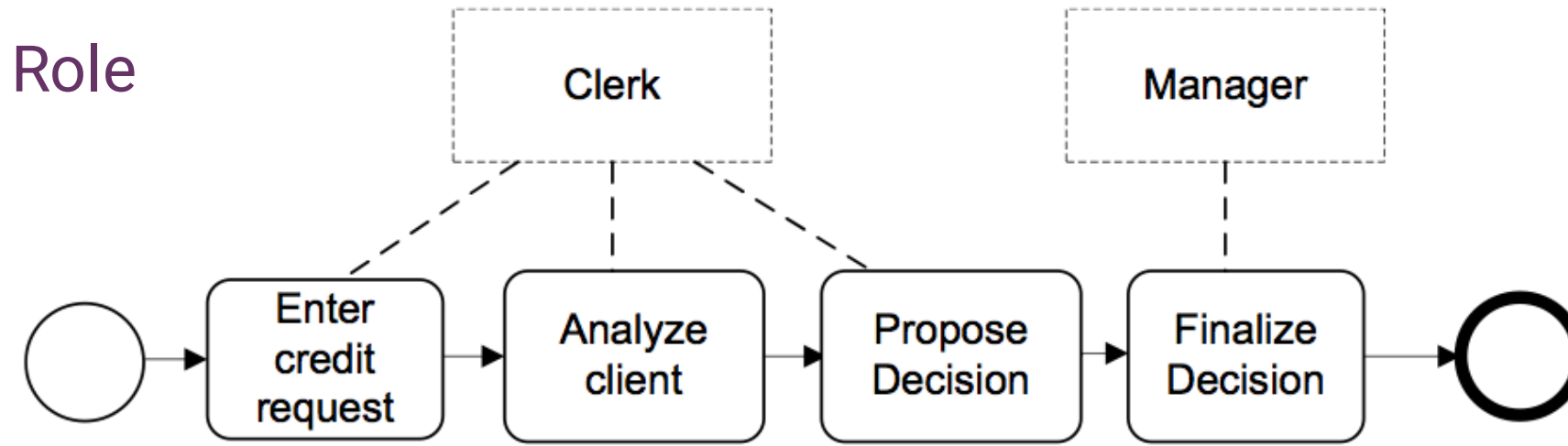


Fig. 3.33. Simple business process model with role information

Organizational Modelling (2/3)

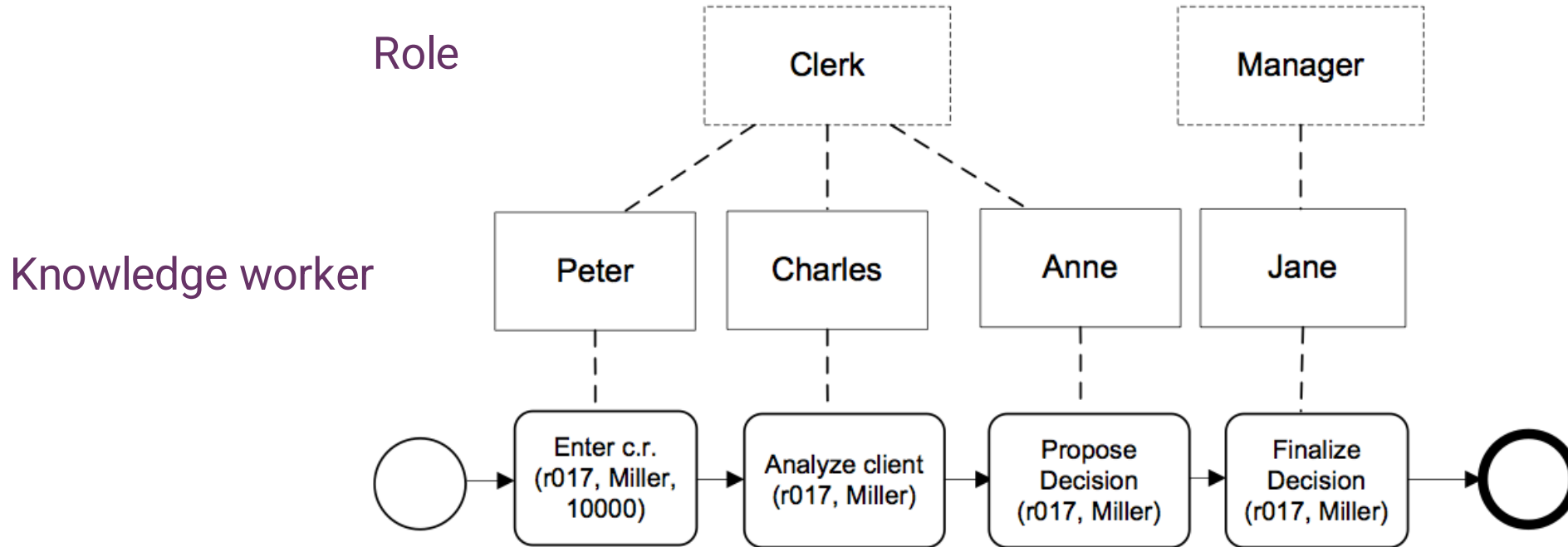


Fig. 3.34. Simple business process instance with knowledge workers associated with activities

- In the example, the hand-over of work from Peter to Charles and from Charles to Anne leads to delays in process executions and should therefore be avoided.

Organizational Modelling (3/3)

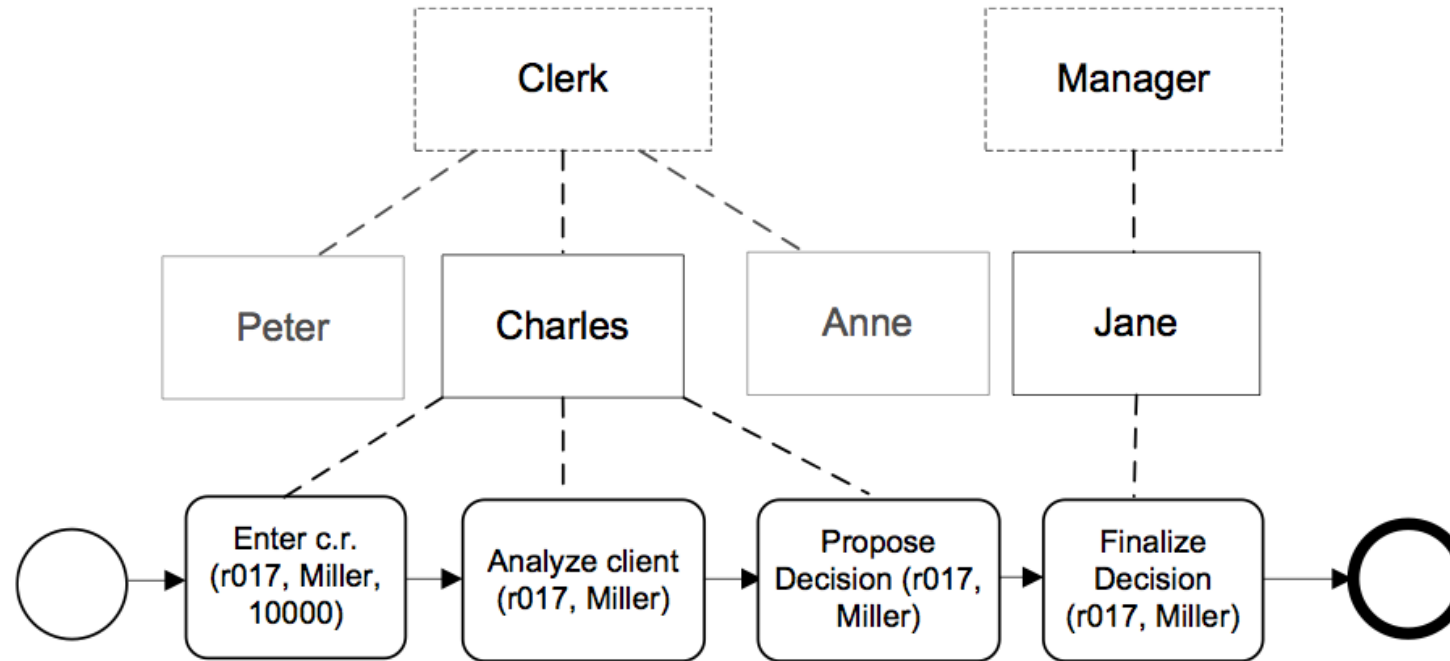
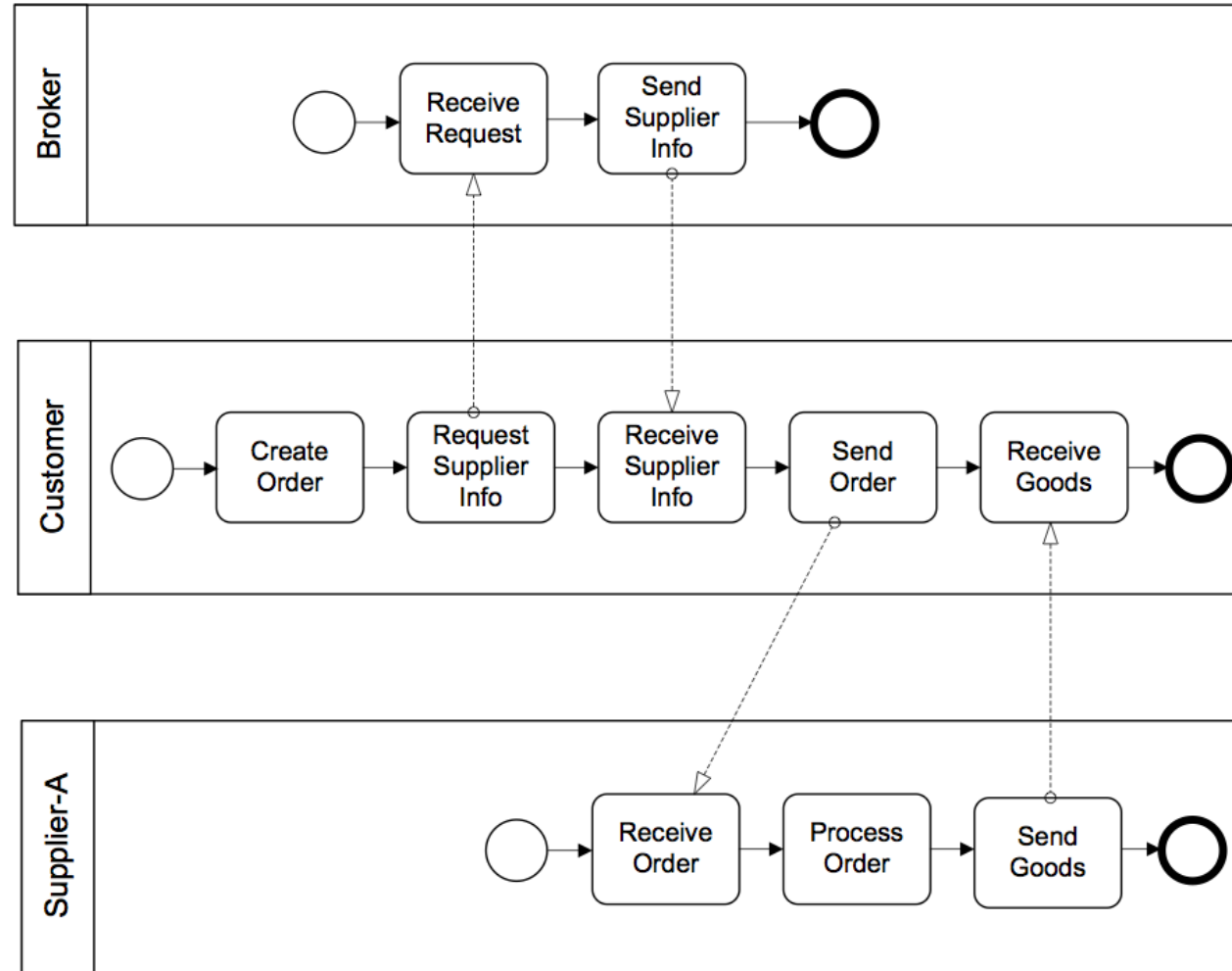


Fig. 3.35. Simple business process instance with one knowledge worker associated with clerk activities

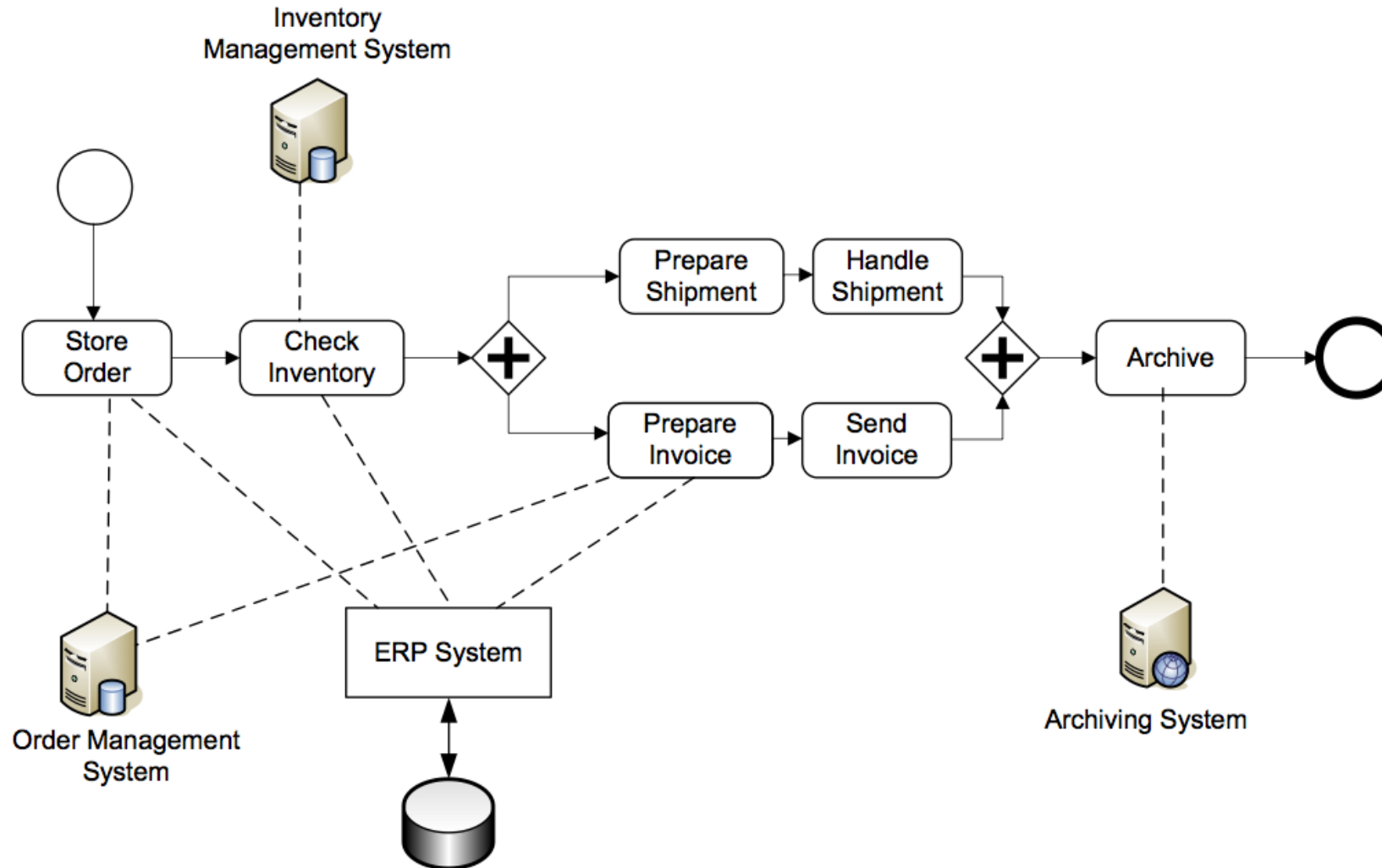
Selection of Business Partners in Process Choreographies



M. Weske: Business Process Management,
 © Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 3.36. Business partner selected at run time, using a broker

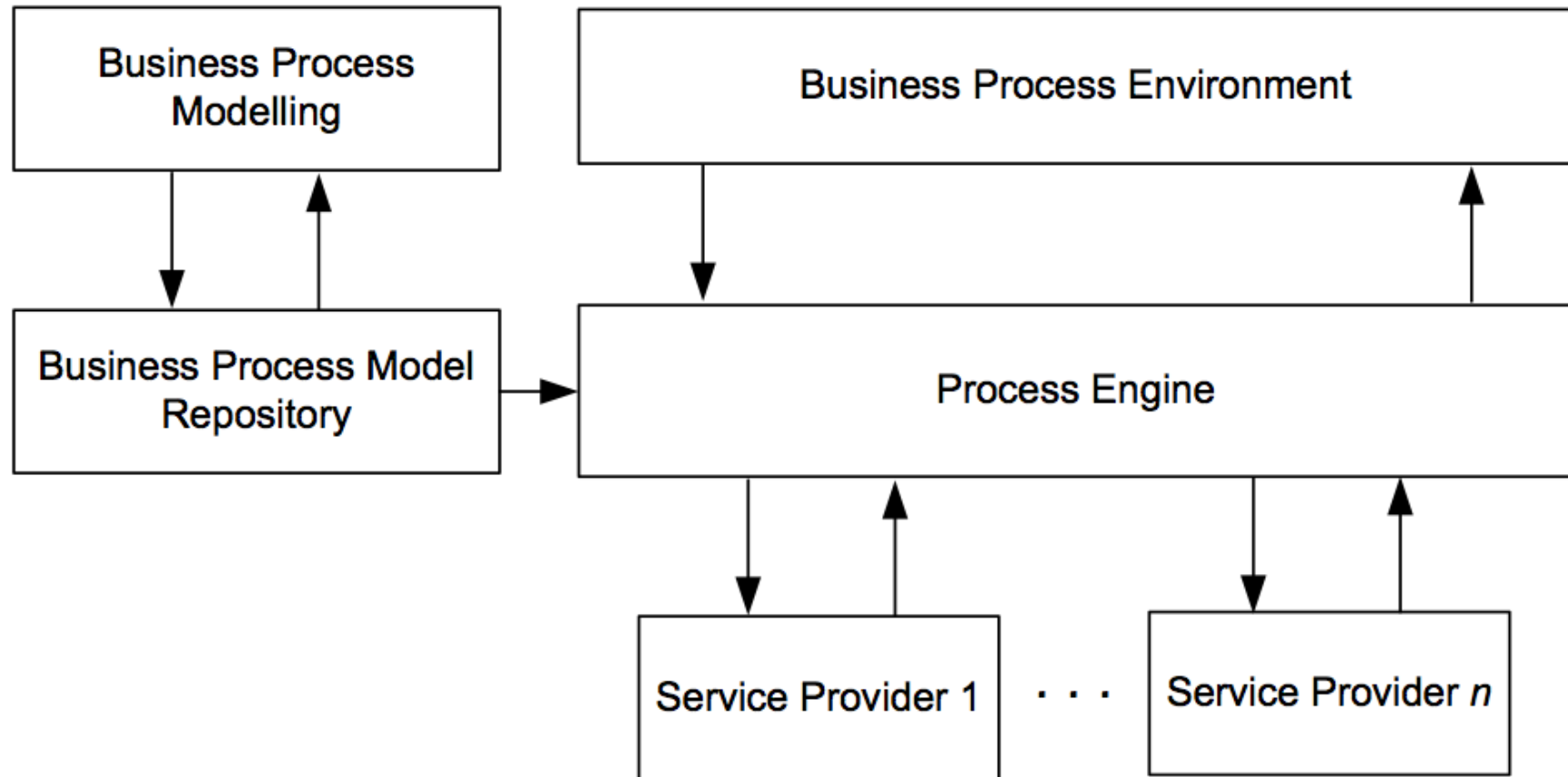
Standardized Software Interfaces



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 3.37. Business process uses ERP systems functionality to realize process activities, while the business process remains unchanged

Architecture of Process Execution Environments



M. Weske: Business Process Management,
© Springer-Verlag Berlin Heidelberg 2012, 2007

Fig. 3.38. Business process management systems architecture model

Mapping the Diagram to Food Delivery Process

Component from Diagram	Food Delivery Example
Business Process Modelling	The company defines a food order process that includes activities like "Customer places order," "Restaurant prepares food," and "Driver delivers food." This process is modeled using BPMN (Business Process Model and Notation)
Business Process Model Repository	Stores predefined order processing workflows . If the company wants to update the process (e.g., adding a "Verify payment" step), the repository holds the updated models.
Business Process Environment	When a customer places an order , the system automatically starts a new process instance based on the defined model.
Process Engine	Executes the process: When an order is placed, the engine moves the process from " Order Received " → " Restaurant Confirmation " → " Food Preparation " → " Delivery ". If an issue arises (e.g., restaurant cancels the order), it triggers an alternative process.
Service Provider 1	Payment System (e.g., Stripe, PayPal) processes the payment when the order is placed.
Service Provider n	Delivery System (e.g., driver dispatch algorithm) assigns the nearest driver

Assignment

- Pick a process from **any company or industry** (e.g., a pizzeria, hospital, bank, online store)
- **Try to draw a process orchestration model** of that company
- Results: Will be discussed
- Group: Form a group of 3
- Time: 15 Minutes
- Why: Familiarize yourself with the topic and PRJ4-BI preparation

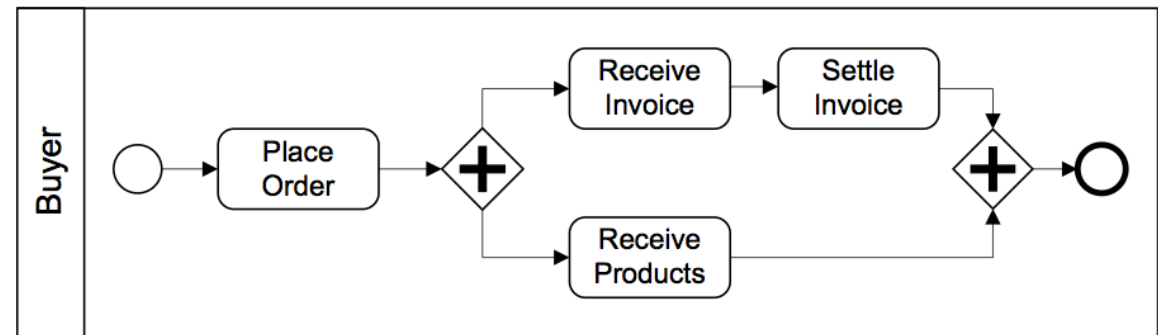


Fig. 1.2. Ordering process of a buyer

Questions?



"Dieses Foto" von Unbekannter Autor ist lizenziert gemäß [CC BY-NC](#)